

ESTIMATING A HURDLE RATE FOR RIO DISNEY

- We **estimated a cost of capital of 6.61%** for the Disney theme park business, using a bottom-up levered beta of 0.7537 for the business.
 - This cost of equity may not adequately reflect the additional risk associated with the theme park being in an emerging market.
 - The concern we would have with using this cost of equity for this project is that it may not adequately reflect the additional risk associated with the theme park being in an emerging market (Brazil).
- We first computed the Brazil country risk premium (by multiplying the default spread for Brazil by the relative equity market volatility) and then re-estimated the cost of equity:
 - Country risk premium for Brazil = $5.5\% + 3\% = 8.5\%$
 - Cost of Equity in US\$ = $2.75\% + 0.7537 (8.5\%) = 9.16\%$
- Using this estimate of the cost of equity, Disney's theme park debt ratio of 10.24% and its after-tax cost of debt of 2.40% (see chapter 4), we can estimate the cost of capital for the project:
 - Cost of Capital in US\$ = $9.16\% (0.8976) + 2.40\% (0.1024) = 8.46\%$

WOULD LEAD US TO CONCLUDE THAT..

- Do not invest in this park. The return on capital of 4.18% is lower than the cost of capital for theme parks of 8.46%; This would suggest that the project should not be taken.
- Given that we have computed the average over an arbitrary period of 10 years, while the theme park itself would have a life greater than 10 years, would you feel comfortable with this conclusion?
 - a. Yes
 - b. No

A TANGENT: FROM NEW TO EXISTING INVESTMENTS: ROC FOR THE ENTIRE FIRM

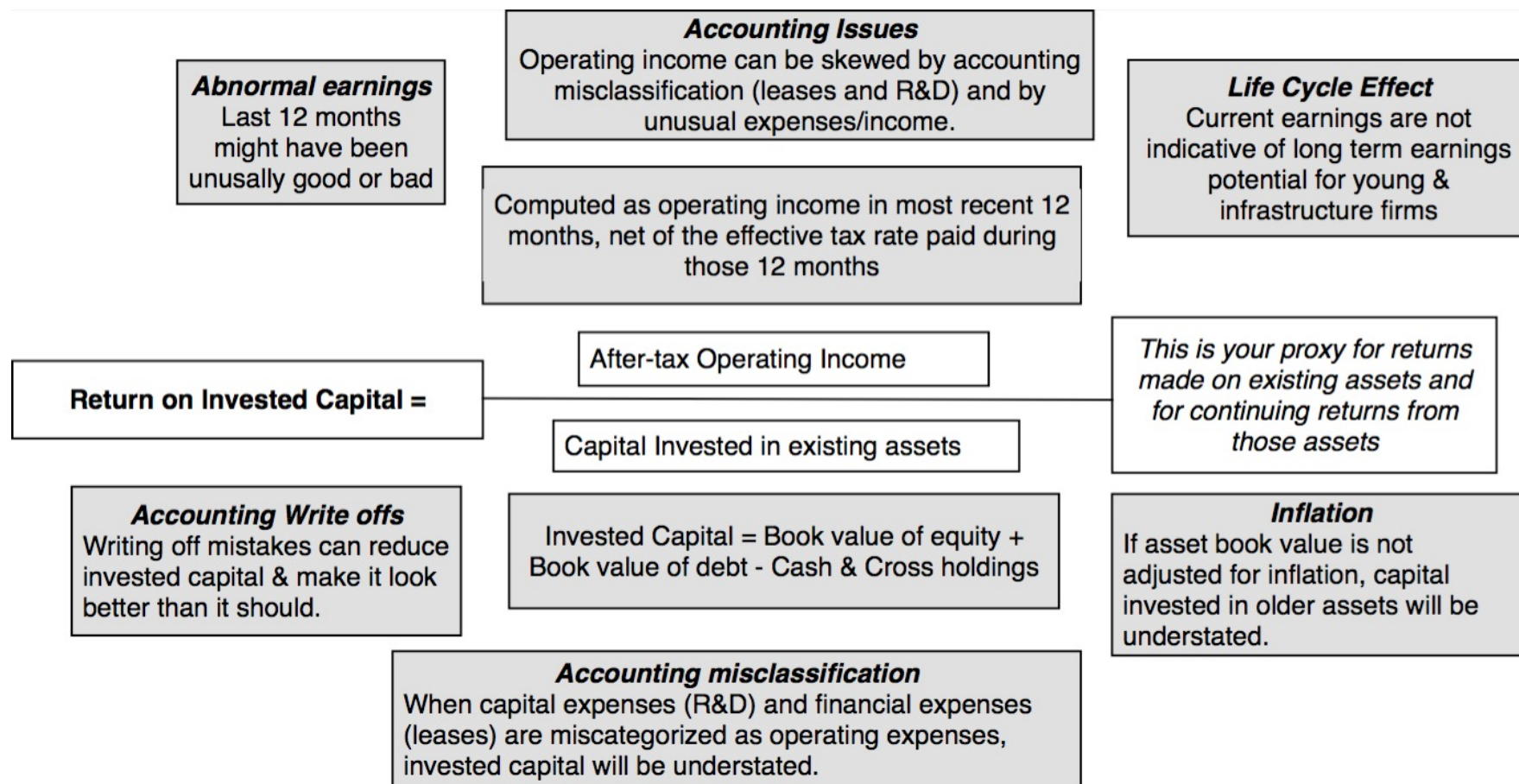
How “good” are the existing investments of the firm?

Assets		Liabilities	
Existing Investments Generate cashflows today Includes long lived (fixed) and short-lived (working capital) assets	Assets in Place	Debt	Fixed Claim on cash flows Little or No role in management <i>Fixed Maturity</i> <i>Tax Deductible</i>
Expected Value that will be created by future investments	Growth Assets	Equity	Residual Claim on cash flows Significant Role in management <i>Perpetual Lives</i>

Measuring ROC for existing investments..

Company	EBIT (1-t)	BV of Debt	BV of Equity	Cash	BV of Capital	Return on Capital	Cost of Capital	ROC - Cost of Capital
Disney	\$6,920	\$16,328	\$41,958	\$3,387	\$54,899	12.61%	7.81%	4.80%
Vale	\$12,432	\$49,246	\$75,974	\$5,818	\$119,402	10.41%	8.20%	2.22%
Baidu	¥9,111	¥13,561	¥27,215	¥10,456	¥30,320	30.05%	12.42%	17.63%
Tata Motors	120,905₹	471,489₹	330,056₹	225,562₹	575,983₹	20.99%	11.44%	9.55%
Bookscape	\$1,775	\$12,136	\$8,250	\$1,250	\$19,136	9.28%	10.30%	-1.02%

THE RETURN ON CAPITAL IS AN ACCOUNTING NUMBER, THOUGH, AND THAT SHOULD SCARE YOU.



RETURN SPREADS GLOBALLY

<i>Region</i>	<i># firms</i>	<i>ROE</i>	<i>COE</i>	<i>% of firms with ROE>COE</i>	<i>ROIC</i>	<i>WACC</i>	<i>% of firms with ROIC>WACC</i>	<i>% of firms with ROIC-WACC>5%</i>	<i>% of firms with ROIC-WACC<5%</i>
Africa and Middle East	2,423	7.55%	10.98%	32.03%	4.77%	9.33%	25.05%	16.59%	83.41%
Australia & NZ	1,798	-12.08%	8.51%	18.19%	-11.59%	8.36%	19.24%	13.68%	86.32%
Canada	2,791	-20.66%	8.64%	11.64%	-18.59%	8.41%	12.54%	8.10%	91.90%
China	7,504	4.34%	10.07%	23.87%	3.36%	8.94%	25.49%	15.27%	84.73%
EU & Environs	5,925	6.73%	9.83%	33.96%	5.48%	8.59%	33.59%	24.76%	75.24%
Eastern Europe & Russia	325	10.17%	10.38%	34.46%	4.32%	9.17%	26.46%	16.31%	83.69%
India	4,446	8.32%	11.12%	34.14%	5.61%	9.90%	29.94%	19.50%	80.50%
Japan	4,020	7.14%	10.05%	33.23%	7.15%	8.62%	41.32%	26.87%	73.13%
Latin America & Caribbean	984	9.28%	12.30%	35.37%	7.37%	9.76%	35.98%	24.19%	75.81%
Small Asia	9,876	5.19%	10.86%	25.65%	3.81%	9.37%	23.78%	14.14%	85.86%
UK	1,125	1.47%	9.71%	29.16%	4.76%	8.74%	37.16%	28.80%	71.20%
United States	6,481	2.64%	8.80%	26.68%	0.05%	7.91%	23.59%	17.74%	82.26%
Global	47,698	4.93%	9.92%	27.54%	3.73%	8.68%	27.12%	18.02%	81.98%

APPLICATION TEST: ASSESSING INVESTMENT QUALITY

- For the most recent period for which you have data, compute the after-tax return on capital earned by your firm, where after-tax return on capital is computed to be
 - $\text{After-tax ROC} = \text{EBIT} (1 - \text{tax rate}) / (\text{BV of debt} + \text{BV of Equity-Cash})_{\text{previous year}}$
- For the most recent period for which you have data, compute the return spread earned by your firm:
 - $\text{Return Spread} = \text{After-tax ROC} - \text{Cost of Capital}$
- Follow up by
 - Examining the reasons why your company earns the excess return (positive or negative) that it does
 - Evaluating whether the company will continue to earn similar excess returns in the future

THE CASH FLOW VIEW OF THIS PROJECT..

	0	1	2	3	4	5	6	7	8	9	10
After-tax Operating Income		-\$32	-\$96	-\$54	\$68	\$202	\$249	\$299	\$352	\$410	\$421
+ Depreciation & Amortization	\$0	\$50	\$425	\$469	\$444	\$372	\$367	\$364	\$364	\$366	\$368
- Capital Expenditures	\$2,500	\$1,000	\$1,188	\$752	\$276	\$258	\$285	\$314	\$330	\$347	\$350
- Change in Non-cash Working Capital		\$0	\$63	\$25	\$38	\$31	\$16	\$17	\$19	\$21	\$5
Cashflow to firm	(\$2,500)	(\$982)	(\$921)	(\$361)	\$198	\$285	\$314	\$332	\$367	\$407	\$434

To get from income to cash flow, we

- I. added back all non-cash charges such as depreciation.
Tax benefits:
- II. subtracted out the capital expenditures
- III. subtracted out the change in non-cash working capital

THE DEPRECIATION TAX BENEFIT

- While depreciation reduces taxable income and taxes, it is a non-cash expense. The benefit of depreciation is therefore the tax benefit. In general, the tax benefit from depreciation can be written as:
 - Tax Benefit = Depreciation * Tax Rate
- Disney Theme Park: Depreciation tax savings (Tax rate = 36.1%)

	1	2	3	4	5	6	7	8	9	10
Depreciation	\$50	\$425	\$469	\$444	\$372	\$367	\$364	\$364	\$366	\$368
Tax Bendfits from Depreciation	\$18	\$153	\$169	\$160	\$134	\$132	\$132	\$132	\$132	\$133

- Proposition 1:** The tax benefit from depreciation and other non-cash charges is greater, the higher your tax rate.
- Proposition 2:** Non-cash charges that are not tax deductible (such as amortization of goodwill) and thus provide no tax benefits have no effect on cash flows.

DEPRECIATION METHODS

- Broadly categorizing, depreciation methods can be classified as straight line or accelerated methods. In straight line depreciation, the capital expense is spread evenly over time, In accelerated depreciation, the capital expense is depreciated more in earlier years and less in later years.
- Assume that you made a large investment this year, and that you are choosing between straight line and accelerated depreciation methods. Which will result in higher net income this year?
 - a. Straight Line Depreciation
 - b. Accelerated Depreciation
- Which will result in higher cash flows this year?
 - a. Straight Line Depreciation
 - b. Accelerated Depreciation

THE CAPITAL EXPENDITURES EFFECT

- Capital expenditures are not treated as accounting expenses but they are cash outflows.
- Capital expenditures can generally be categorized into two groups
 - **New (or Growth) capital expenditures** are capital expenditures designed to create new assets and future growth
 - **Maintenance capital expenditures** refer to capital expenditures designed to keep existing assets.
 - Both initial and maintenance capital expenditures reduce cash flows
- The **need for maintenance capital expenditures will increase with the life of the project**. In other words, a 25-year project will require more maintenance capital expenditures than a 2-year project.

TO CAP EX OR NOT TO CAP EX?

- Assume that you run a movie/TV streaming business, and that you have an expense this year of \$ 100 million from acquiring new subscribers, by offering them special deals and free access. Your accountant tells you that you can expense this item or capitalize and depreciate it over three years. Which will have a more positive effect on income?
 - a. Expense it
 - b. Capitalize and Depreciate it
- Which will have a more positive effect on cash flows?
 - a. Expense it
 - b. Capitalize and Depreciate it

THE WORKING CAPITAL EFFECT

- Intuitively, money invested in inventory or in accounts receivable cannot be used elsewhere. It, thus, represents a drain on cash flows
- To the degree that some of these investments can be financed using supplier credit (accounts payable), the cash flow drain is reduced.
- Investments in working capital are thus cash outflows
 - Any increase in working capital reduces cash flows in that year
 - Any decrease in working capital increases cash flows in that year
- To provide closure, working capital investments need to be salvaged at the end of the project life.
- **Proposition 1:** The failure to consider working capital in a capital budgeting project will overstate cash flows on that project and make it look more attractive than it really is.
- **Proposition 2:** Other things held equal, a reduction in working capital requirements will increase the cash flows on all projects for a firm.

THE INCREMENTAL CASH FLOWS ON THE PROJECT

	0	1	2	3	4	5	6	7	8	9	10
After-tax Operating Income		-\$32	-\$96	-\$54	\$68	\$202	\$249	\$299	\$352	\$410	\$421
+ Depreciation & Amortization	\$0	\$50	\$425	\$469	\$444	\$372	\$367	\$364	\$364	\$366	\$368
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- Change in non-cash Working Capital		\$0	\$63	\$25	\$38	\$31	\$16	\$17	\$19	\$21	\$5
Cashflow to firm	(\$2,500)	(\$982)	(\$921)	(\$361)	\$198	\$285	\$314	\$332	\$367	\$407	\$434
+ Pre-project investment (sunk)	\$500										
- Pre-project Depreciation * tax rate		\$18	\$18	\$18	\$18	\$18	\$18	\$18	\$18	\$18	\$18
+ Non-incremental Allocated Expense (1-t)		\$0	\$80	\$112	\$160	\$200	\$220	\$242	\$266	\$292	\$298
Incremental Cash flow to the firm	(\$2,000)	(\$1,000)	(\$860)	(\$267)	\$340	\$467	\$516	\$555	\$615	\$681	\$715

\$ 500 million has already been spent & \$ 50 million in depreciation will exist anyway

2/3rd of allocated G&A is fixed.
Add back this amount (1-t)
Tax rate = 36.1%

A MORE DIRECT WAY OF GETTING TO INCREMENTAL CASH FLOWS

	0	1	2	3	4	5	6	7	8	9	10
Revenues		\$0	\$1,250	\$1,750	\$2,500	\$3,125	\$3,438	\$3,781	\$4,159	\$4,575	\$4,667
Direct Expenses		\$0	\$788	\$1,103	\$1,575	\$1,969	\$2,166	\$2,382	\$2,620	\$2,882	\$2,940
Incremental Depreciation		\$0	\$375	\$419	\$394	\$322	\$317	\$314	\$314	\$316	\$318
Incremental G&A		\$0	\$63	\$88	\$125	\$156	\$172	\$189	\$208	\$229	\$233
Incremental Operating Income		\$0	\$25	\$141	\$406	\$678	\$783	\$896	\$1,017	\$1,148	\$1,175
- Taxes		\$0	\$9	\$51	\$147	\$245	\$283	\$323	\$367	\$415	\$424
Incremental after-tax Operating income		\$0	\$16	\$90	\$260	\$433	\$500	\$572	\$650	\$734	\$751
+ Incremental Depreciation		\$0	\$375	\$419	\$394	\$322	\$317	\$314	\$314	\$316	\$318
- Capital Expenditures	\$2,000	\$1,000	\$1,188	\$752	\$276	\$258	\$285	\$314	\$330	\$347	\$350
- Change in non-cash Working Capital		\$0	\$63	\$25	\$38	\$31	\$16	\$17	\$19	\$21	\$5
Cashflow to firm	(\$2,000)	(\$1,000)	(\$859)	(\$267)	\$340	\$466	\$516	\$555	\$615	\$681	\$715

SUNK COSTS

- **What is a sunk cost?** Any expenditure that has already been incurred and cannot be recovered (even if a project is rejected) is called a sunk cost.
 - A test market for a consumer product and R&D expenses for a drug (for a pharmaceutical company) would be good examples.
- **The sunk cost rule:** When analyzing a project, sunk costs should not be considered since they are not incremental.
- **A Behavioral Aside:** It is a well-established finding in psychological and behavioral research that managers find it almost impossible to ignore sunk costs.

TEST MARKETING AND R&D: THE QUANDARY OF SUNK COSTS

- A consumer product company has spent \$ 100 million on test marketing. Looking at only the incremental cash flows (and ignoring the test marketing), the project looks like it will create \$25 million in value for the company. Should it take the investment?
 - a. Yes
 - b. No
- Now assume that every investment that this company has shares the same characteristics (Sunk costs > Value Added). The firm will clearly not be able to survive. What is the solution to this problem?

ALLOCATED COSTS

- Firms allocate costs to individual projects from a centralized pool (such as general and administrative expenses) based upon some characteristic of the project (sales is a common choice, as is earnings)
 - For large firms, these allocated costs can be significant and result in the rejection of projects
 - To the degree that these costs are not incremental (and would exist anyway), this makes the firm worse off.
- Thus, it is only the incremental component of allocated costs that should show up in project analysis.

BREAKING OUT G&A COSTS INTO FIXED AND VARIABLE COMPONENTS: A SIMPLE EXAMPLE

- Assume that you have a time series of revenues and G&A costs for a company.

Year	Revenues	G&A Costs
1	\$1,000	\$250
2	\$1,200	\$270
3	\$1,500	\$300

- What percentage of the G&A cost is variable?

TO TIME-WEIGHTED CASH FLOWS

- Incremental cash flows in the earlier years are worth more than incremental cash flows in later years.
- In fact, cash flows across time cannot be added up. They have to be brought to the same point in time before aggregation.
- This process of moving cash flows through time is
 - discounting, when future cash flows are brought to the present
 - compounding, when present cash flows are taken to the future

PRESENT VALUE MECHANICS

■ Cash Flow Type Discounting Formula Compounding Formula

■ 1. Simple CF
 $CF_n / (1+r)^n$
 $CF_0 (1+r)^n$

■ 2. Annuity

$$A \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right]$$

$$A \left[\frac{(1+r)^n - 1}{r} \right]$$

■ 3. Growing Annuity

$$A(1+g) \left[\frac{1 - \frac{(1+g)^n}{(1+r)^n}}{r-g} \right]$$

■ 4. Perpetuity
 A/r

■ 5. Growing Perpetuity
Expected Cashflow next year/(r-g)

DISCOUNTED CASH FLOW MEASURES OF RETURN

- **Net Present Value (NPV):** The net present value is the sum of the present values of all cash flows from the project (including initial investment).
 - NPV = Sum of the present values of all cash flows on the project, including the initial investment, with the cash flows being discounted at the appropriate hurdle rate (cost of capital, if cash flow is cash flow to the firm, and cost of equity, if cash flow is to equity investors)
 - Decision Rule: Accept if $NPV > 0$
- **Internal Rate of Return (IRR):** The internal rate of return is the discount rate that makes the net present value equal zero.
 - It is the percentage rate of return, based upon incremental time-weighted cash flows.
 - Decision Rule: Accept if $IRR > \text{hurdle rate}$

CLOSURE ON CASH FLOWS

- **Salvage Value:** In a project with a finite and short life, you would need to compute a salvage value, which is the expected proceeds from selling all of the investment in the project at the end of the project life. It is often set equal to book value of fixed assets and working capital
- **Terminal Value:** In a project with an infinite or very long life, we compute cash flows for a reasonable period, and then compute a terminal value for this project, which is the present value of all cash flows that occur after the estimation period ends..
- Assuming the project lasts forever, and that cash flows after year 10 grow 2% (the inflation rate) forever, the present value at the end of year 10 of cash flows after that can be written as:
 - Terminal Value in year 10 = $CF \text{ in year } 11 / (\text{Cost of Capital} - \text{Growth Rate}) = 715 (1.02) / (.0846 - .02) = \$ 11,275 \text{ million}$

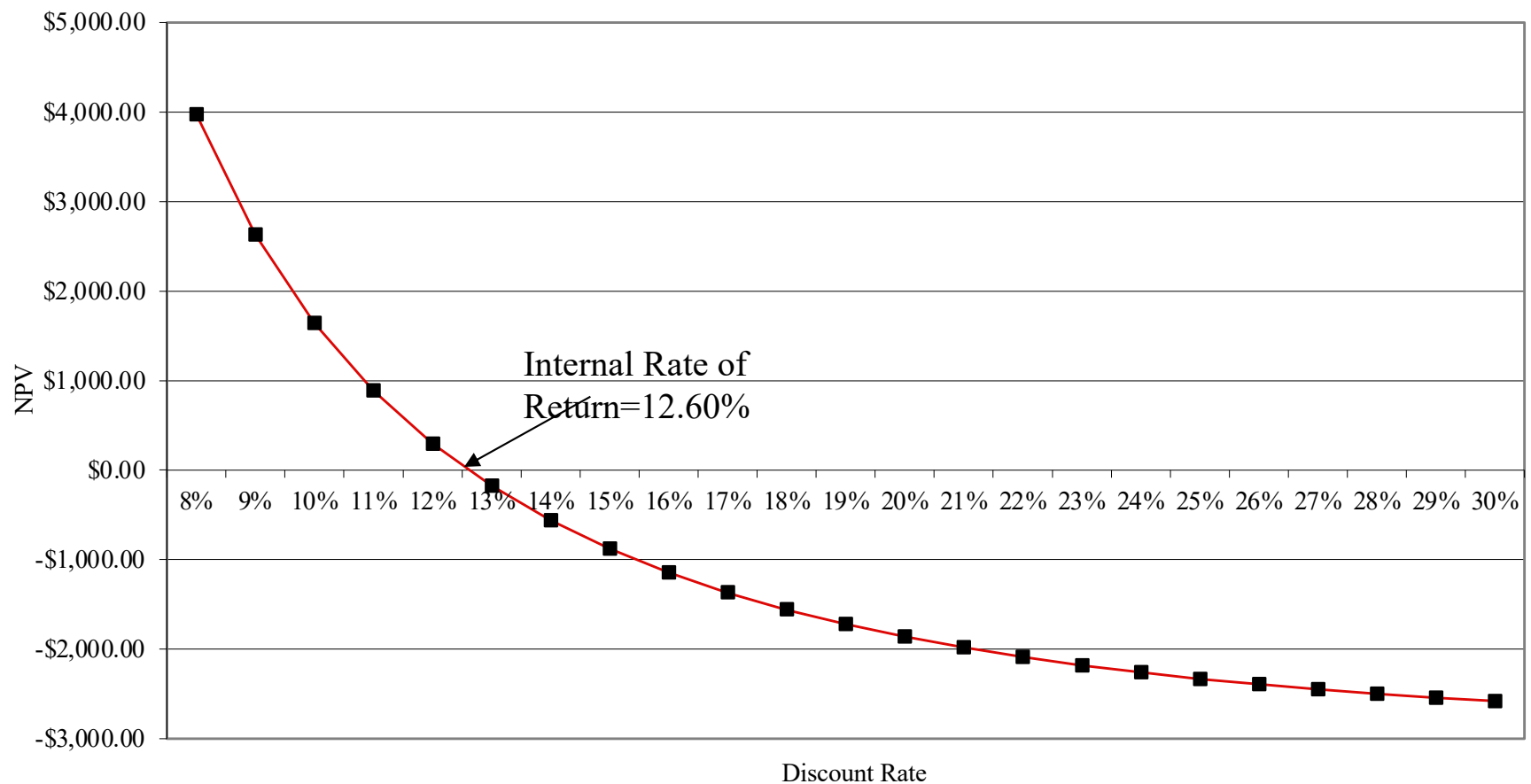
WHICH YIELDS A NPV OF..

Year	Annual Cashflow	Terminal Value	Present Value
0	-\$2,000		-\$2,000
1	-\$1,000		-\$922
2	-\$859		-\$730
3	-\$267		-\$210
4	\$340		\$246
5	\$466		\$311
6	\$516		\$317
7	\$555		\$314
8	\$615		\$321
9	\$681		\$328
10	\$715	\$11,275	\$5,321
			\$3,296

WHICH MAKES THE ARGUMENT THAT..

- The project should be accepted. The positive net present value suggests that the project will add value to the firm and earn a return in excess of the cost of capital.
 - By taking the project, Disney will increase its value as a firm by \$3,296 million.
 - Will the market price increase by the same amount? Why or why not?

THE IRR OF THIS PROJECT



THE IRR SUGGESTS..

- **The project is a good one.** Using time-weighted, incremental cash flows, this project provides a return of 12.60%. This is greater than the cost of capital of 8.46%.
- **The IRR and the NPV will yield similar results most of the time**, though there are differences between the two approaches that may cause project rankings to vary depending upon the approach used. They can yield different results, especially when comparing across projects because
 - A project can have **only one NPV**, whereas it can have more than one IRR.
 - The **NPV is a dollar surplus value**, whereas the IRR is a percentage measure of return. The NPV is therefore likely to be larger for “large scale” projects, while the IRR is higher for “small-scale” projects.
 - The NPV assumes that **intermediate cash flows get reinvested at the “hurdle rate”**, which is based upon what you can make on investments of comparable risk, while the IRR assumes that intermediate cash flows get reinvested at the “IRR”.

DOES THE CURRENCY MATTER?

- The Rio Disney Theme Park analysis was done in dollars, even though the project will be based in Brazil and will have cash flows primarily in Brazilian Reais.
- Would your assessment of the project have been any different if we had done the analysis in Brazilian Reais?
 - a. Yes
 - b. No

THE “CONSISTENCY RULE” FOR CASH FLOWS

- The cash flows on a project and the discount rate used should be defined in the same terms.
 - If cash flows are in dollars (\$R), the discount rate has to be a dollar (\$R) discount rate
 - If the cash flows are nominal (real), the discount rate has to be nominal (real).
- If consistency is maintained, the project conclusions should be identical, no matter what cash flows are used.

DISNEY THEME PARK: PROJECT ANALYSIS IN \$R

- The inflation rates were assumed to be 9% in Brazil and 2% in the United States. The \$R/dollar rate at the time of the analysis was 2.35 \$R/dollar.
- The expected exchange rate was derived assuming purchasing power parity.
 - Expected Exchange Rate_t = Exchange Rate today * (1.09/1.02)^t
- The expected growth rate after year 10 is still expected to be the inflation rate, but it is the 9% \$R inflation rate.
- The cost of capital in \$R was derived from the cost of capital in dollars and the differences in inflation rates:
 - $$\text{\$R Cost of Capital} = (1 + \text{US \$ Cost of Capital}) \frac{(1 + \text{Exp Inflation}_{\text{Brazil}})}{(1 + \text{Exp Inflation}_{\text{US}})} - 1$$
 - $$= (1.0846) (1.09/1.02) - 1 = 15.91\%$$

DISNEY THEME PARK: \$R NPV

Expected Exchange Rate_t
= Exchange Rate today * (1.09/1.02)^t

Discount at \$R cost of capital
= (1.0846) (1.09/1.02) – 1 = 15.91%

Year	Cashflow (\$)	\$R/\$	Cashflow (\$R)	Present Value
0	-R\$ 2,000.00	R\$ 2.35	-R\$ 4,700.00	-R\$ 4,700.00
1	-R\$ 1,000.00	R\$ 2.51	-R\$ 2,511.27	-R\$ 2,166.62
2	-R\$ 859.03	R\$ 2.68	-R\$ 2,305.29	-R\$ 1,715.95
3	-R\$ 267.39	R\$ 2.87	-R\$ 766.82	-R\$ 492.45
4	R\$ 340.22	R\$ 3.06	R\$ 1,042.63	R\$ 577.68
5	R\$ 466.33	R\$ 3.27	R\$ 1,527.21	R\$ 730.03
6	R\$ 516.42	R\$ 3.50	R\$ 1,807.31	R\$ 745.36
7	R\$ 555.08	R\$ 3.74	R\$ 2,075.89	R\$ 738.63
8	R\$ 614.95	R\$ 4.00	R\$ 2,457.65	R\$ 754.45
9	R\$ 681.46	R\$ 4.27	R\$ 2,910.36	R\$ 770.81
10	R\$ 11,989.85	R\$ 4.56	R\$ 54,719.84	R\$ 12,503.50
				R\$ 7,745.43

NPV = R\$ 7,745/2.35= \$ 3,296 Million

NPV is equal to NPV in dollar terms